

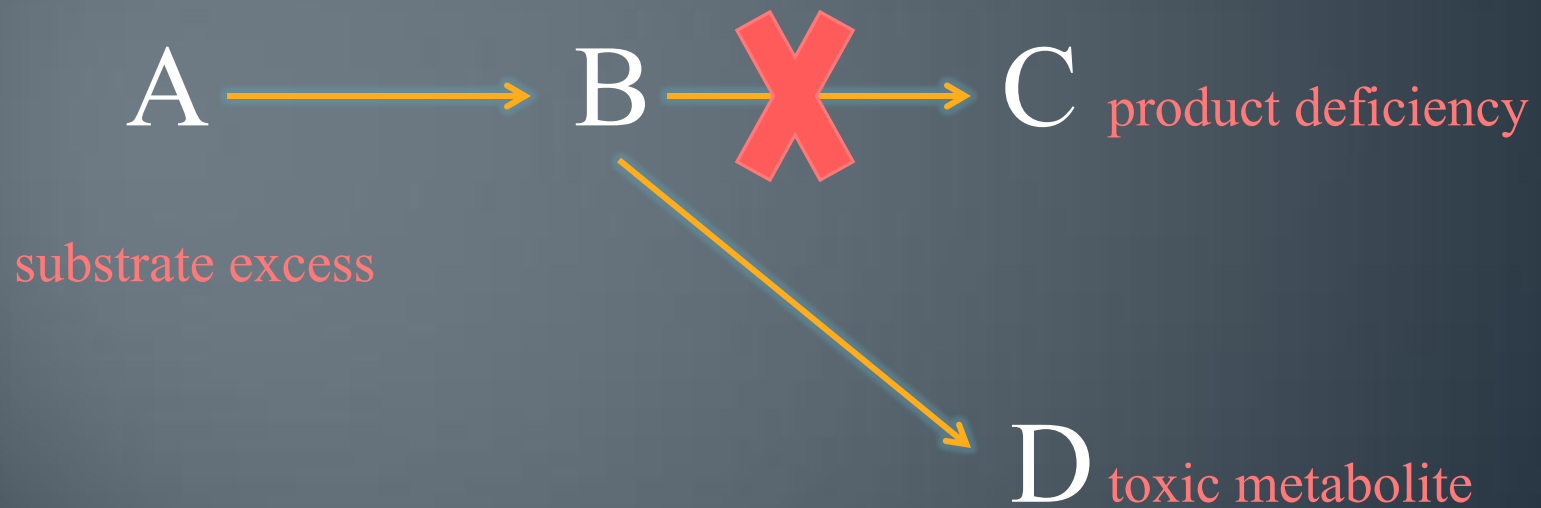
Inborn errors of metabolism

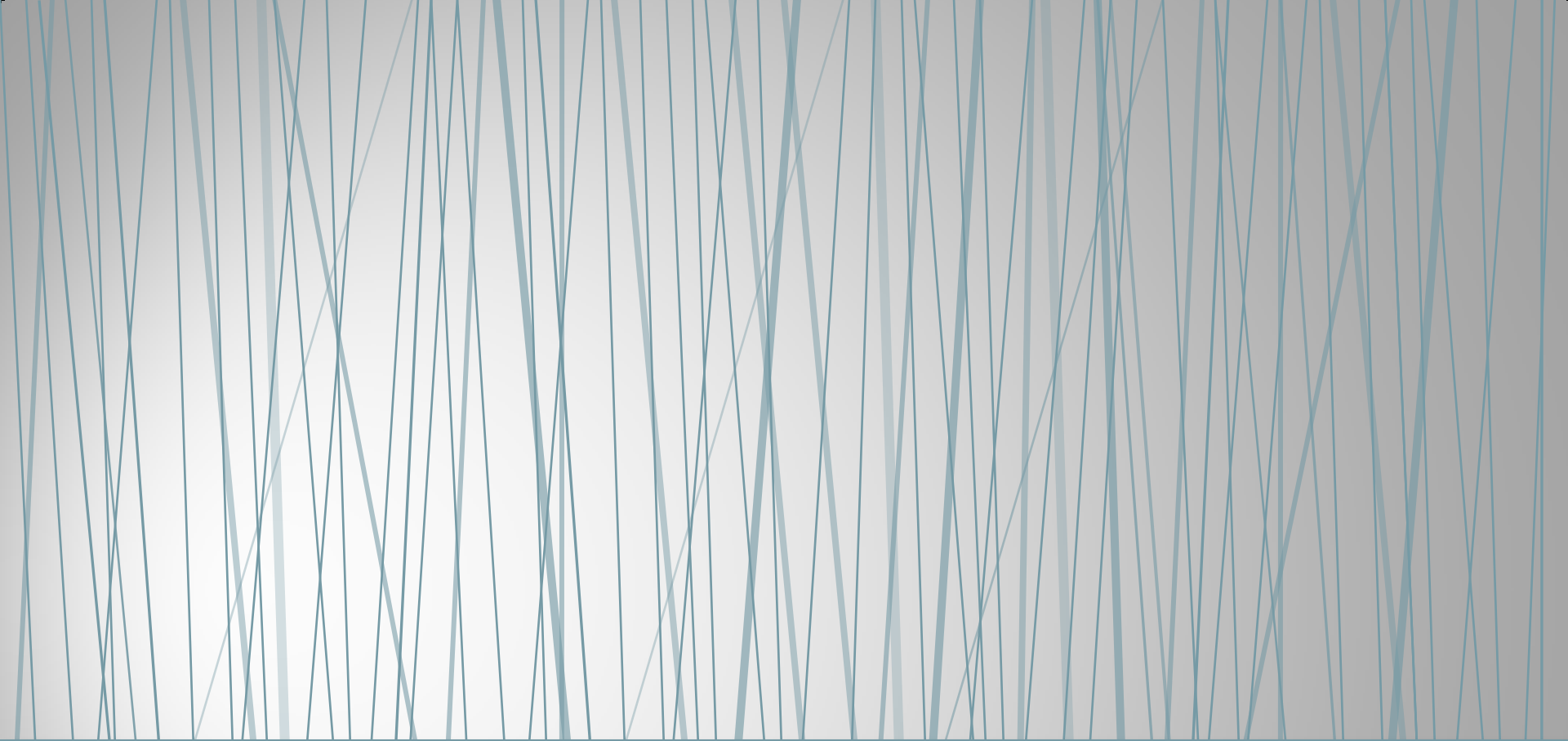
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- Inborn errors of metabolism comprise a large class of genetic diseases involving disorders of metabolism.
- The majority are due to defects of single genes that code for enzymes that facilitate conversion of various substances (substrates) into others (products).
- In most of the disorders, problems arise due to accumulation of substances which are toxic or interfere with normal function, or to the effects of reduced ability to synthesize essential compounds.

Inborn errors of metabolism are now often referred to as **congenital metabolic diseases** or **inherited metabolic diseases**

Garrod's hypothesis



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Major categories of inherited metabolic diseases

Major categories of inherited metabolic diseases (1)

Disorders of carbohydrate metabolism

- E.g., glycogen storage disease

Disorders of amino acid metabolism

- E.g., phenylketonuria , maple syrup urine disease, glutaric acidemia type 1

Disorders of organic acid metabolism (organic acidurias)

- E.g., alcaptonuria

Disorders of fatty acid oxidation and mitochondrial metabolism

- E.g., medium chain acyl dehydrogenase deficiency (glutaric acidemia type 2)

Disorders of porphyrin metabolism

- E.g., acute intermittent porphyria

Major categories of inherited metabolic diseases (2)

Disorders of purine or pyrimidine metabolism

- E.g., Lesch-Nyhan syndrome

Disorders of steroid metabolism

- E.g., congenital adrenal hyperplasia

Disorders of mitochondrial function

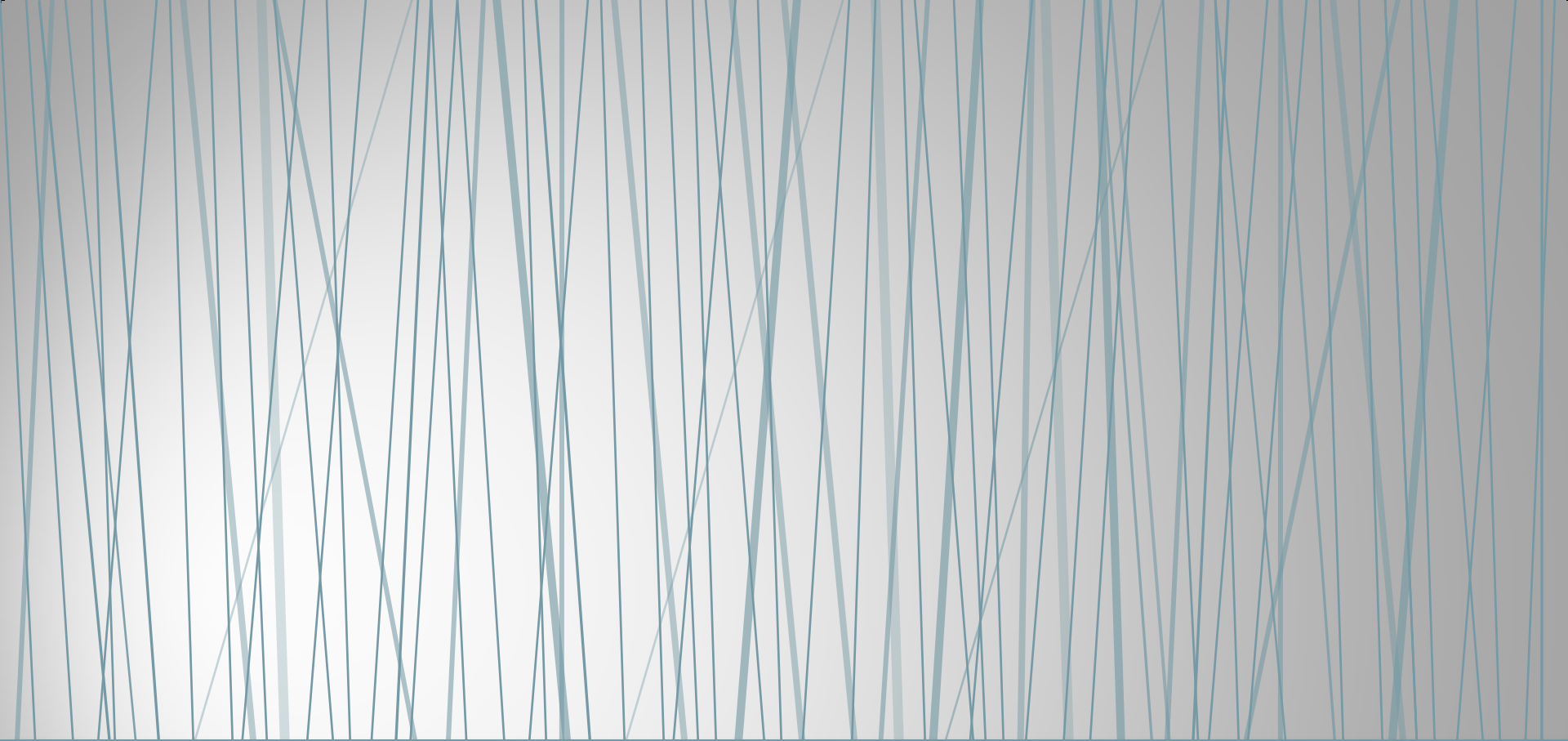
- E.g., Kearns-Sayre syndrome

Disorders of peroxisomal function

- E.g., Zellweger syndrome

Lysosomal storage disorders

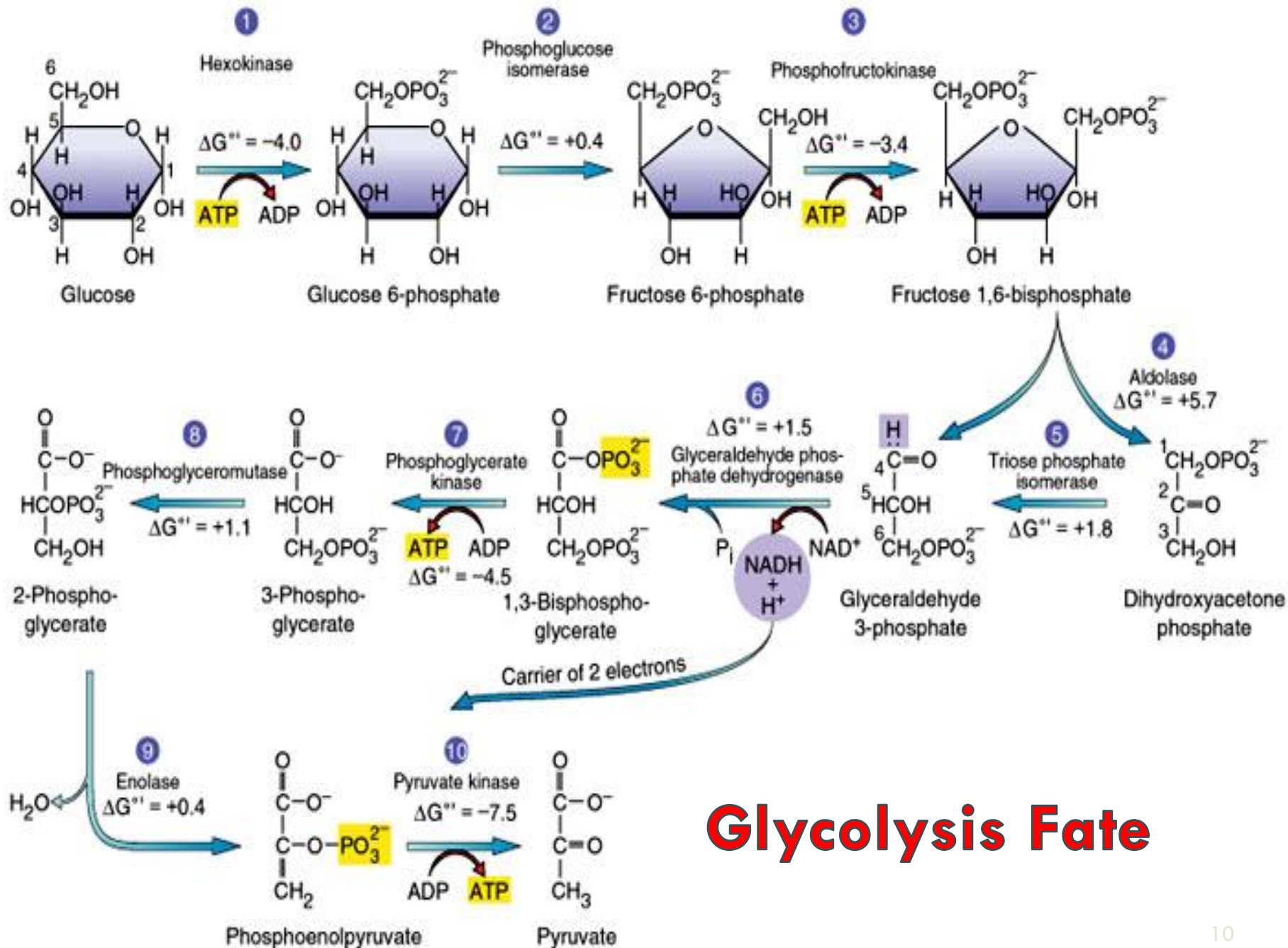
- E.g., Gaucher's disease

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Disorders of carbohydrate metabolism, Glycogen storage disease

Glycogen storage disease

- **Glycogen storage disease** (GSD, also glycogenosis and dextrinosis) is the result of defects in the processing of glycogen synthesis or breakdown within muscles, liver, and other cell types. GSD has two classes of cause: genetic and acquired.
- **Genetic GSD** is caused by any inborn error of metabolism (genetically defective enzymes) involved in these processes.



* GSD Type I (Von Gierke's disease)

- **Symptoms:**

Hypoglycemia, Hyperlipidemia, Hepatomegaly, Lactic acidosis, and Hyperuricemia.

- **Progression:** Growth failure

- **Enzyme deficiency:** (**glucose-6-phosphatase**) which is an enzyme that hydrolyzes glucose-6-phosphate resulting in the creation of a phosphate group and free glucose. **This deficiency impairs the ability of the liver to produce free glucose from glycogen and from gluconeogenesis. Since these are the two principal metabolic mechanisms by which the liver supplies glucose to the rest of the body during periods of fasting, it causes severe hypoglycemia.**

* GSD Type I (Von Gierke's disease)

Treatment:

- The essential treatment goal is prevention of hypoglycemia and the secondary metabolic derangements by frequent feedings of foods high in glucose or starch (which is readily digested to glucose). To compensate for the inability of the liver to provide sugar, the total amount of dietary carbohydrate should approximate the 24-hour glucose production rate. The diet should contain approximately 65-70% carbohydrate, 10-15% protein, and 20-25% fat. At least a third of the carbohydrates should be supplied through the night, so that a young child goes no more than 3–4 hours without carbohydrate intake
- Two methods have been used to achieve this goal in young children: (1) continuous nocturnal gastric infusion of glucose or starch; and (2) night-time feedings of uncooked cornstarch.

** GSD Type II (Pompe's disease)

- Is an autosomal recessive metabolic disorder, which damages muscle and nerve cells throughout the body. It is caused by an accumulation of glycogen in the lysosome due to deficiency of the **lysosomal acid alpha-glucosidase** enzyme that transforms glycogen into glucose in lysosomes.
- The build-up of glycogen causes progressive muscle weakness (myopathy) throughout the body and affects various body tissues, particularly in the heart, skeletal muscles, and weakness facial and oral muscles. Pompe's disease is one of the infiltrative causes of restrictive cardiomyopathy and hepatomegaly.
- caused by a mutation in a gene (acid alpha-glucosidase: also known as acid maltase) on long arm of chromosome 17.

**** GSD Type II (Pompe's disease)**

Nutrition & Weight Maintenance

- Because of weakened facial and oral muscles, patients of all ages, from infants to adults, may experience difficulties eating. Trouble with sucking, chewing, and/or swallowing can lead to insufficient caloric intake, problems maintaining a healthy weight, and a general failure to thrive. Inadequate nutrition may even lead to endogenous muscle protein breakdown.

Several approaches can address these issues:

- Physical therapy to help strengthen muscles and allow for independent feeding.
- Modification of food texture to facilitate swallowing and reduce the risk of aspiration.
- Carefully balanced diets to maximize nutrients and provide protein to muscles.
- Tube feeding, most commonly in severely ill infants.

** GSD Type II (Pompe's disease)

Treatment:

- In 2006, the European Medicines Agency (EMA) and the U.S. Food and Drug Administration (FDA) both granted marketing approval for the drug **Myozyme (alglucosidase alfa)** for treatment of Pompe disease. Myozyme replaces the enzyme missing in the disease, which helps break down glucose.
- Early diagnosis and early treatment leads to much better outcomes.
- **Progression:** Death by age ~2 years.

*** GSD Type V (McArdle Disease)

- Is a metabolic disorder, caused by a deficiency of enzyme **Myophosphorylase**, which is the muscle isoform of the enzyme glycogen phosphorylase.
- This enzyme helps break down glycogen into **glucose-1-phosphate**, so that it can be utilized within the muscle cell.
- **Symptoms:** The onset of this disease is usually noticed in childhood, but often not diagnosed until the third or fourth decade of life. Symptoms include exercise intolerance with myalgia, early fatigue, painful cramps, weakness of exercising muscles and myoglobinuria. Myoglobinuria, the condition where myoglobin is present in urine, may result from serious damage to the muscles, or **rhabdomyolysis**, where skeletal muscle cells breakdown rapidly, sending their contents into the bloodstream.

*** GSD Type V (McArdle Disease)

Treatment/Therapy

- Oral **vitamin B₆** appears to impart greater resistance to fatigue. No specific therapy exists, but combined aerobic exercise programs and high-protein diets may help. Some patients learn the limits of their exercise and work within their restrictions, going on to live fairly normal lives.
- Supervised exercise programs have been recommended to lessen the risks of extended inactivity.
- Sucrose treatment is now being recommended prior to exercise.
- **Progression:** Renal failure due to myoglobinuria.

*** GSD Type V (McArdle Disease)

Vit B6 rich food	High proteins food
Spinach	Soy protein isolate
Red bell peppers	Gelatin
Garlic	Egg, white
Carrots	Fish meat
Peas	Milk
Potatoes	Chicken
Milk	Nuts
Egg	Peanut butter
Fish	Steak
Liver	Cheese
Meat (red)	Hamburger
Broccoli	Broccoli

**** GSD Type VII (Tarui's Disease)

- Is metabolic disorder with autosomal recessive inheritance
Phosphofructokinase deficiency.

Pathophysiology:

- In this condition, a deficiency phosphofructokinase enzyme impairs the ability of cells such as erythrocytes and skeletal muscles to use carbohydrates for energy.
- The mutation impairs the ability of phosphofructokinase to phosphorylate fructose-6-phosphate prior to its cleavage into glyceraldehyde which enters the Krebs cycle, effectively limiting energy production.
- Unlike most other GSD, it directly affects glycolysis.

**** GSD Type VII (Tarui's Disease)

Presentation

- The disease presents with exercise-induced muscle cramps and weakness (sometimes rhabdomyolysis), myoglobinuria, as well as with **haemolytic anaemia** causing dark urine. Hyperuricemia is common. Phosphofructokinase deficiency also presents in a rare infantile form, results in severe myopathy and leads to death in the infancy or early childhood.

Treatment/interventions

- There is no cure for Tarui disease, but various treatments may alleviate symptoms and complications.
- Individuals with Tarui disease should be observant to myoglobinuria, presenting as a dark discoloration of the urine. Owing to the risk of kidney damage, medical help should be sought **immediately** if symptoms arise. Dialysis is needed if toxic waste products accumulate owing to renal failure (uraemia).

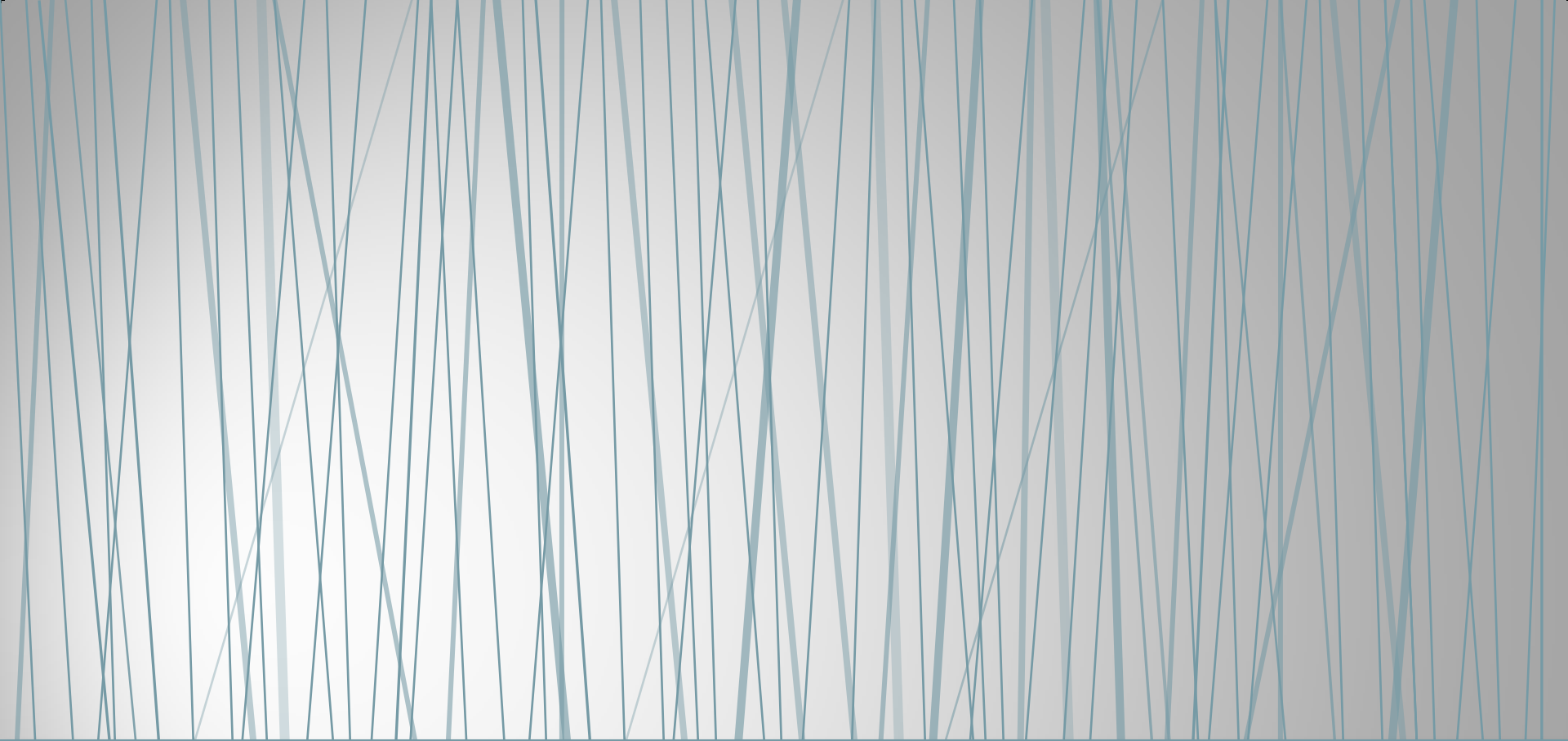
**** GSD Type VII (Tarui's Disease)

Treatment/interventions

- In Tarui's disease, jaundice is mild and generally does not require treatment.
- High uric acid concentrations that may cause gout can be treated with drugs which lower uric acid levels in the blood.
- The effectiveness of dietary management remains unclear. It is possible that food with a high fat content (notably fatty fish) has a beneficial effect, as the glycerol in neutral fat can replace glucose as a source of energy. It may be possible to "teach" the skeletal muscle cells to oxidise fatty acids rather than glucose to produce energy.
- Individuals with Tarui's disease should avoid intensive muscle activity that has many negative consequences for physical and mental health.

Other Types of Glycogen Storage Diseases

Type of GSD	Eponym	Enzyme deficiency	Progression and Complications
GSD III	Cori's or Forbes disease	Glycogen debrancher	Hypoglycemia and myopathy
GSD IV	Andersen disease	Glycogen branching Enzyme	Liver cirrhosis, death at age ~5 years
GSD VI	Hers disease	Liver glycogen phosphorylase	Hypoglycemia and Hepatomegaly
GSD IX		Phosphorylase kinase	Delayed motor development, Growth retardation
GSD XI	Fanconi-Bickel syndrome	Glucose transporter, GLUT2	Hypoglycemia and Hepatomegaly
GSD XII	Red Cell Aldolase	Aldolase A	Exercise intolerance, and muscle cramps
GSD XIII		B-enolase	Exercise intolerance, and muscle cramps
GSD O		Glycogen synthase	Hypoglycemia

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Disorders of amino acid metabolism

Phenylketonuria (PKU)

- (PKU) is an autosomal recessive metabolic genetic disorder characterized by a deficiency in the hepatic enzyme **phenylalanine hydroxylase (PAH)**. This enzyme is necessary to metabolize the phenylalanine (Phe) to the tyrosine. When PAH is deficient, phenylalanine accumulates and is converted into phenylpyruvate, which is detected in the urine.
- It can cause problems with brain development, leading to progressive mental retardation, brain damage, and seizures.
- Optimal treatment involves lowering blood (Phe) levels to a safe range and monitoring diet and cognitive development.
- PKU is normally detected using the HPLC test after birth.

Phenylketonuria (PKU)

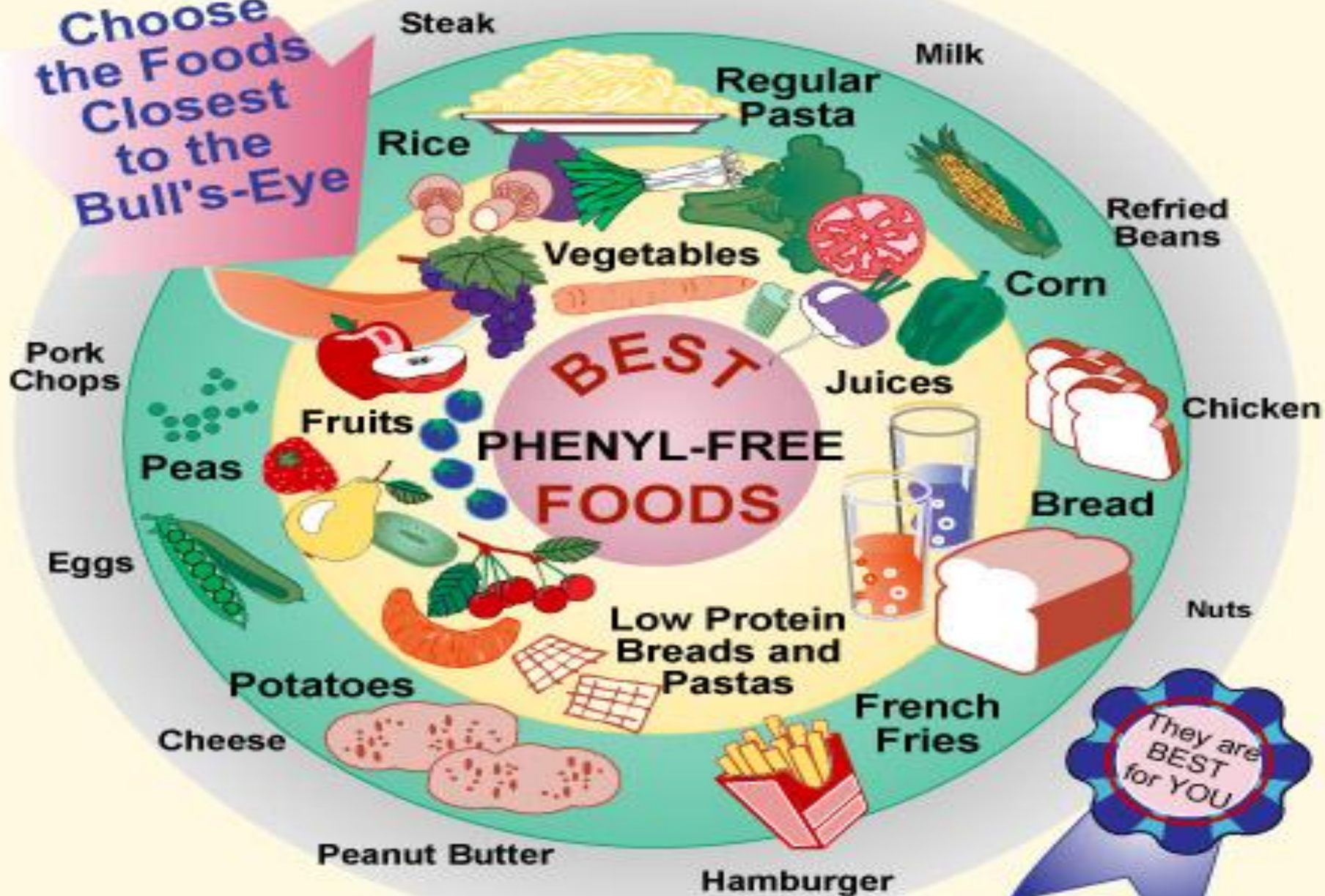
Signs and Symptoms:

- the disease may present clinically with seizures, albinism (excessively fair hair and skin), and a "musty odor" to the baby's sweat and urine (due to phenylacetate, one of the ketones produced).
- **Treatment:** by managing and controlling (Phe) levels through diet, or a combination of diet and medication.
- All PKU patients must adhere to a special diet low in phenylalanine for at least the first 16 years of their lives. This requires severely restricting or eliminating foods high in phenylalanine, such as meat, chicken, fish, eggs, nuts, cheese, legumes, cow milk and other dairy products. Starchy foods such as potatoes, bread, pasta, and corn must be monitored.
- Infants require a commercial formula of milk that free from (Phe).

Phenylketonuria (PKU)

- Tyrosine, which is normally derived from phenylalanine, must be supplemented.
- The sweetener of **aspartame** must be avoided, as aspartame consists of two amino acids: phenylalanine and aspartic acid.
- The oral administration of tetrahydrobiopterin (or BH₄) (a cofactor for the oxidation of phenylalanine) can reduce blood levels of this amino acid in certain patients.
- For childhood, we can add some fruits and vegetables the low in (Phe) which provide essential vitamins and minerals.

Choose
the Foods
Closest
to the
Bull's-Eye



Maple syrup urine disease (MSUD)

- Also called **branched-chain ketoaciduria**, is an autosomal recessive metabolic disorder affecting branched-chain amino acids. It is one type of organic acidemia.
- MSUD is caused by a deficiency of the **branched-chain alpha-keto acid dehydrogenase** complex (BCKDH), leading to a buildup of the branched-chain amino acids (**leucine**, **isoleucine**, and **valine**) and their toxic by-products in the blood and urine.
- The disease is characterized in an infant by the presence of sweet-smelling urine, with an odor similar to that of maple syrup. Infants with this disease seem healthy at birth but if left untreated suffer severe brain damage and eventually die.
- From early infancy, symptoms of the condition include poor feeding, vomiting, dehydration, lethargy, seizures, hypoglycaemia, ketoacidosis, pancreatitis, coma and neurological decline.

Maple syrup urine disease (MSUD)

Management:

- Keeping MSUD under control requires careful monitoring of blood chemistry and involves both a special diet and frequent testing.
- A diet with minimal levels of the amino acids leucine, isoleucine, and valine must be maintained in order to prevent neurological damage. As these three amino acids are required for proper metabolic function in all people, specialized protein preparations containing substitutes and adjusted levels of the amino acids have been synthesized and tested, allowing MSUD patients to meet normal nutritional requirements without causing harm.

Maple syrup urine disease (MSUD)

Leucine (Food)	Isoleucine (Food)	Valine (Food)
Soybeans Lentils Cowpea اللوبيا Beef (lean and trimmed) Peanuts Salmon fish Shrimp Nuts Eggs	Eggs Soy protein Seeweed Milk Cheese Sesame seeds Sunflower seeds Cod liver	Closed to Isoleucine sources

Glutaric aciduria type 1

- **Glutaric acidemia type 1** (or "Glutaric Aciduria", "GA1", or "GAT1") is an inherited disorder in which the body is unable to break down completely the amino acids **lysine**, **hydroxylysine** and **tryptophan**. Excessive levels of their intermediate breakdown products (**glutaric acid**, **glutaryl-CoA**, **3-hydroxyglutaric acid**, **glutaconic acid**) can accumulate and cause damage to the brain (and also other organs), but particularly the basal ganglia, which are regions that help regulate movement. GA1 causes secondary carnitine deficiency, as glutaric acid, like other organic acids, is detoxified by carnitine. Mental retardation may also occur.

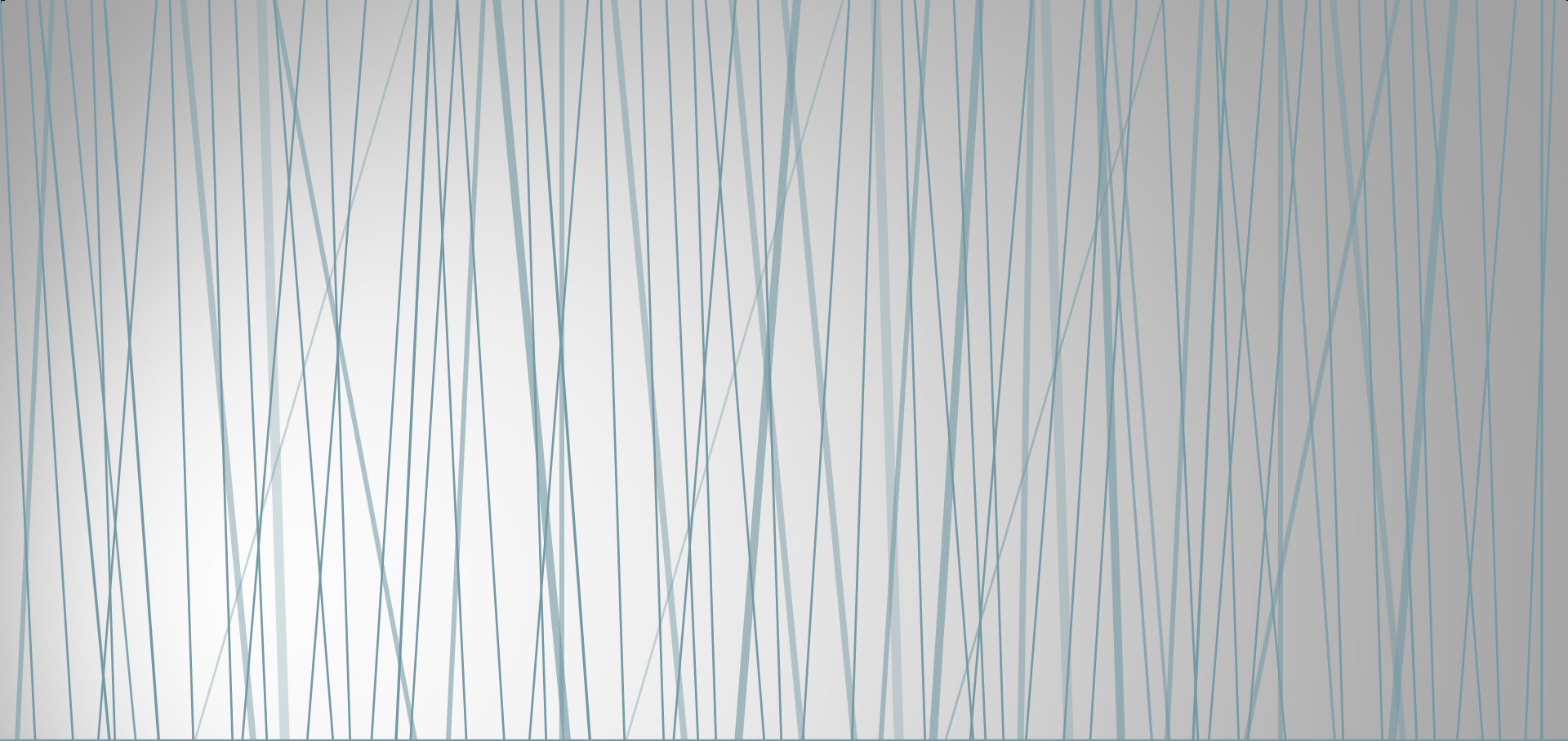
Treatment of: **Glutaric aciduria type 1**

- Correction of secondary **Carnitine** depletion by oral supplementation.
- **Precursor restriction:** Dietary control may help limit progression of the neurological damage.
- The entry of tryptophan to the brain is crucial in the proper synthesis of the neurotransmitter serotonin in the brain
5-hydroxytryptophan.
- The precursor of serotonin that is not metabolized to glutaryl-CoA, glutaric acid and secondary metabolites, could be used as an adjunct to selective tryptophan restriction.

Carnitine sources

<i>Food of animal origin (uncooked)</i>	<i>L-Carnitine [mg/100g]</i>	<i>Food of plant origin (uncooked)</i>	<i>L-Carnitine [mg/100g]</i>
Lamb	190	Mushroom	2.6
Beef	143	Carrot	0.4
Pork	25	Bread	0.4
Poultry	13	Rice	0.3
Fish	3-10	Banana	0.1
Egg	0.8	Tomato	0.1

Table 1: Plant foods contain only traces of L-Carnitine while foods of animal origin contain higher amounts of this nutrient.

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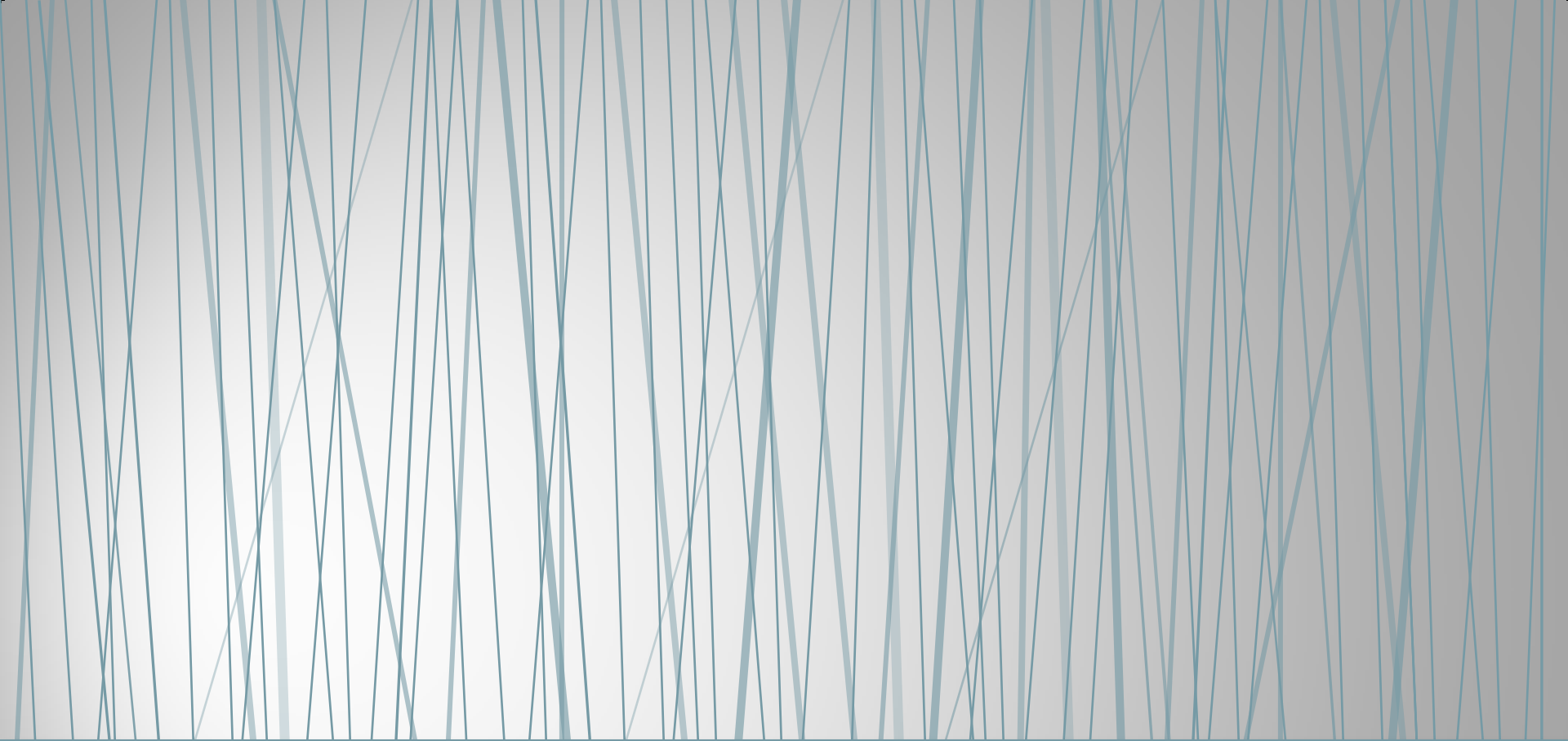
Disorders of organic acid metabolism (organic acidurias)

Alkaptonuria

- **Alkaptonuria (black urine disease)** is a rare inherited genetic disorder of phenylalanine and tyrosine metabolism. This is an autosomal recessive condition that is due to a defect in the enzyme **homogentisate 1,2-dioxygenase**, which participates in the degradation of tyrosine.
- As a result, a toxic tyrosine byproduct called **homogentisic acid** (or alkapton) accumulates in the blood and is excreted in urine in large amounts. Excessive homogentisic acid causes damage to cartilage (leading to osteoarthritis) and heart valves as well as precipitating as kidney stones.

Alkaptonuria

- No treatment modality has been demonstrated to reduce the complications of alkaptonuria.
- Commonly recommended treatments include dietary restriction of phenylalanine and tyrosine and large doses of **ascorbic acid** (vitamin C).
- Dietary restriction may be effective in children, but benefits in adults have not been demonstrated.

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Disorders of fatty acid oxidation and mitochondrial metabolism

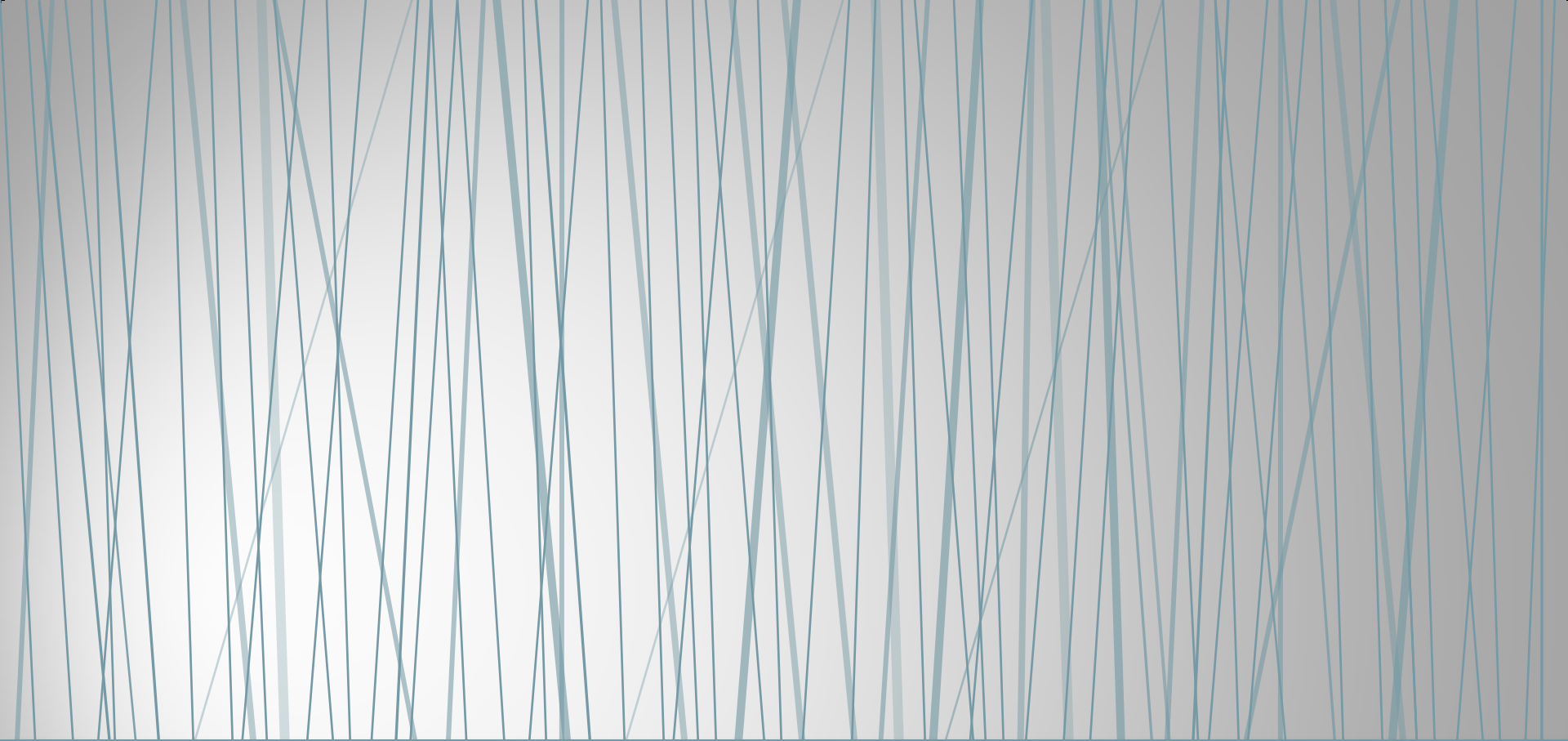
Medium chain acyl dehydrogenase deficiency (glutaric acidemia type 2)

- **Medium-chain acyl-coenzyme A dehydrogenase deficiency** (MCADD) is a fatty acid oxidation disorder associated with inborn errors of metabolism. It is due to defects in the enzyme complex known as *medium-chain acyl dehydrogenase* (MCAD) and reduced activity of this complex. This complex oxidizes medium chain fatty acids (Fatty acids having 6-12 carbons) while reducing FAD to FADH_2 .
- It is recognized as one of the more rare causes of sudden infant death syndrome (SIDS).

Medium chain acyl dehydrogenase deficiency (glutaric acidemia type 2)

Treatment:

- There is no cure for MCADD, but once diagnosed, adverse effects can be prevented by proper management.
- The most important part of treatment is to ensure that patients never go without food for longer than 10–12 hours (overnight fast).
- Patients with an illness causing loss of appetite or severe vomiting may need IV glucose to make sure that the body is not dependent on fatty acids for energy. Patients also usually adhere to a low-fat diet.
- Patients may also take daily doses of **carnitine**, which helps reduce toxic accumulation of fatty acids by forming acyl carnitines, which are excreted in the urine.
- Severity of symptoms seems to decrease after puberty.

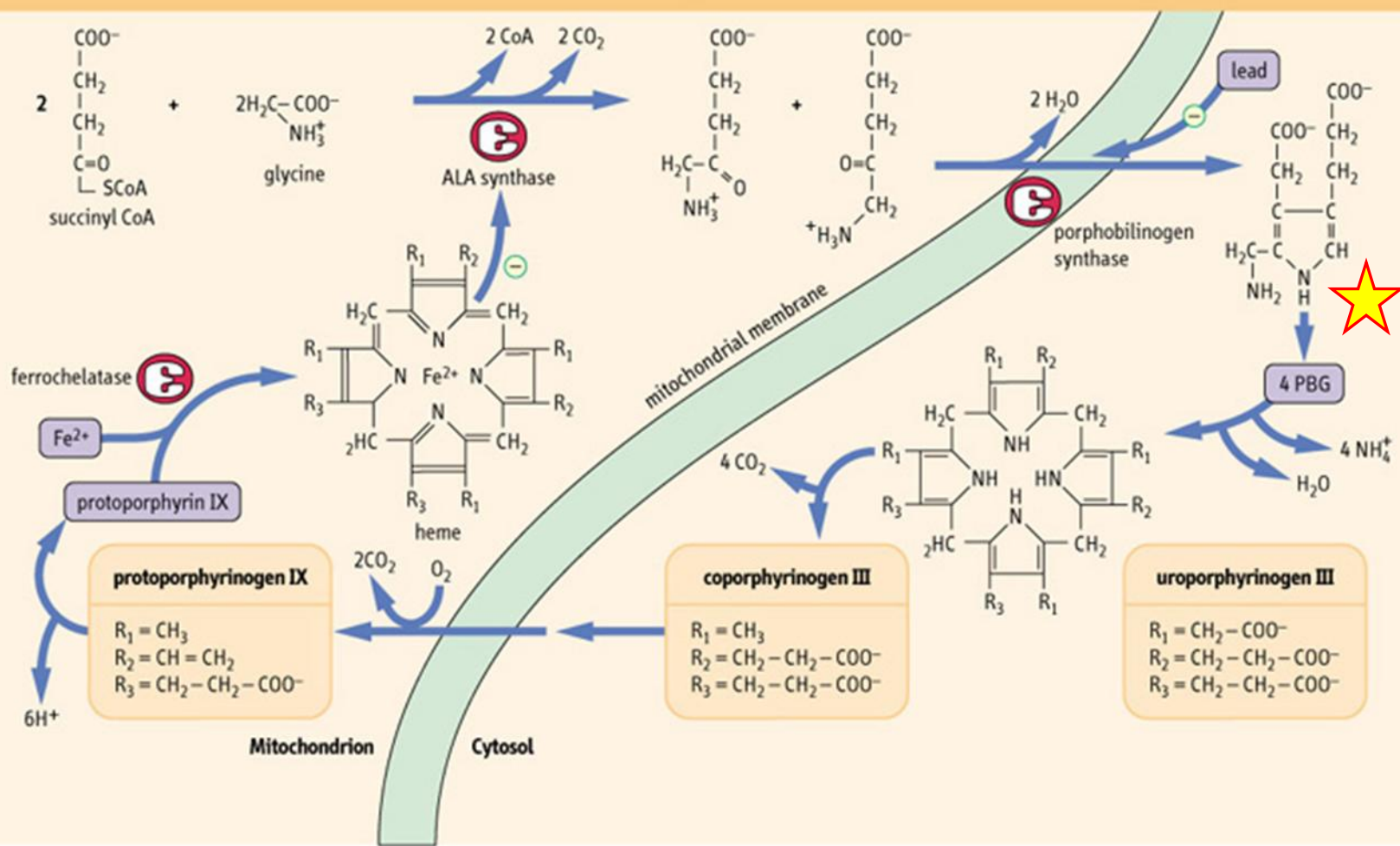
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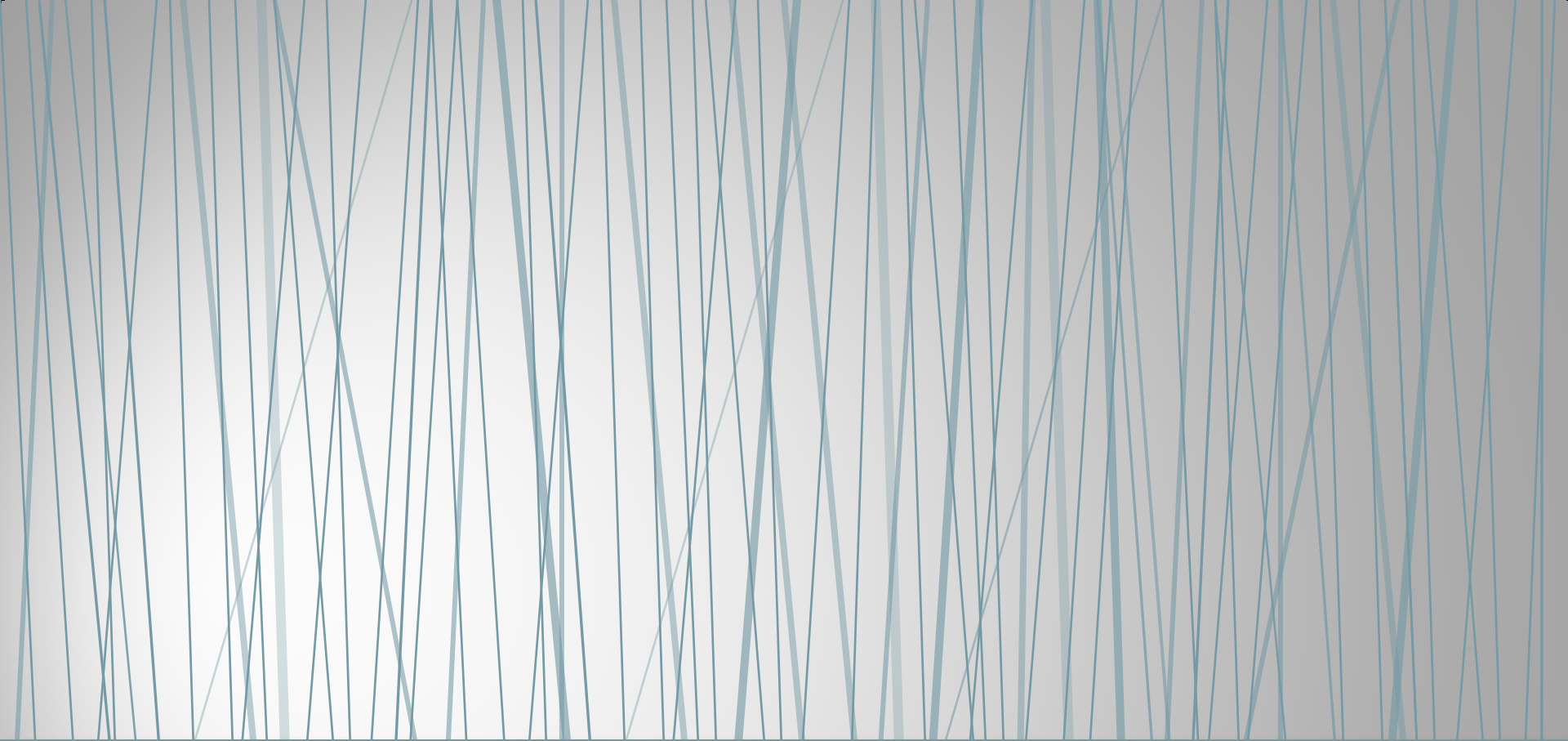
Disorders of porphyrin metabolism

Acute intermittent porphyria

- **Acute intermittent porphyria (AIP)** is a rare autosomal dominant metabolic disorder affecting the production of **heme**, the oxygen-binding prosthetic group of hemoglobin. It is characterized by a deficiency of the enzyme **porphobilinogen deaminase**.
- Symptoms of AIP include abdominal pain, constipation, muscle weakness, and also tend to develop various psychiatric illnesses.
- **Treatment:** A high-carbohydrate or glucose 10% infusion is recommended, which may aid in recovery.
- Iron intake should be adequate to avoid iron deficiency.

Heme synthesis



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Disorders of purine or pyrimidine metabolism

Lesch-Nyhan Syndrome (Hyperuricemias)

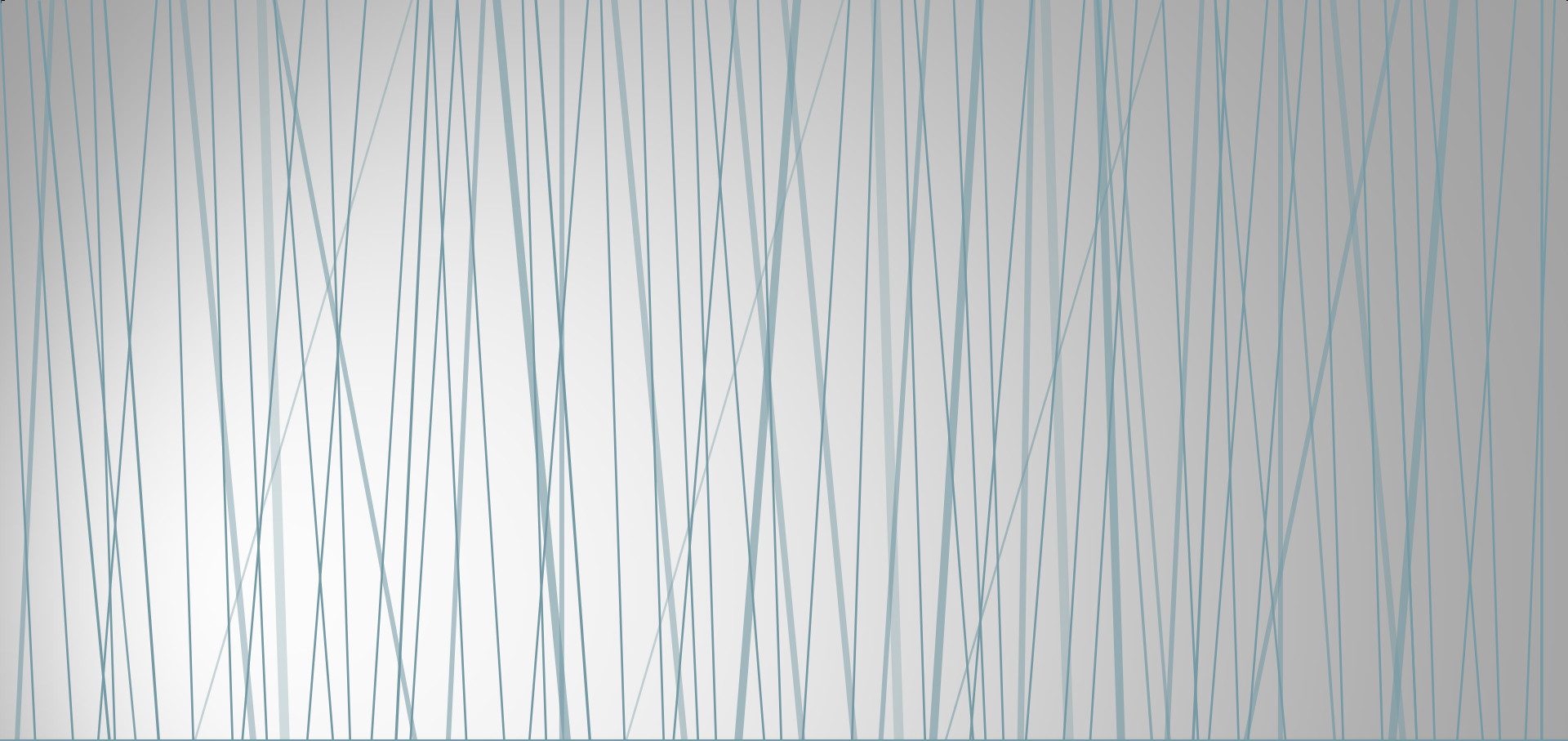
- Is inherited (X-linked recessive) disorder caused by a deficiency of the **hypoxanthine-guanine phosphoribosyltransferase enzyme** (HGPRT), produced by mutations in the HPRT gene.
- The HGPRT deficiency causes a build-up of uric acid in all body fluids. This results in both **hyperuricemia** and **hyperuricosuria**, associated with:
 - 1- Severe gout and kidney problems,
 - 2- Neurological signs include poor muscle control,
 - 3- Moderate mental retardation.
- These complications usually appear in the first year of life.

Lesch-Nyhan Syndrome (LNS)

- In the second year of life, a particularly striking feature of LNS is self-mutilating behaviors, characterized by lip and finger biting.
- The LNS should associated with teeth extraction and restrains to avoid self-mutilating behaviors.

Treatment:

- The elevated level of uric acid in blood and urine doesn't relate to high purine diet, but due to physiological error.
- Because a lack of HGPRT causes the body to poorly utilize vitamin B₁₂, some boys may develop megaloblastic anemia and neurological symptoms.

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Disorders of steroid metabolism

Congenital adrenal hyperplasia

- **Congenital adrenal hyperplasia (CAH)** refers to any of several autosomal recessive diseases resulting from mutations of genes for enzymes mediating the biochemical steps of production of cortisol from cholesterol by the adrenal glands (steroidogenesis).
- Most of these conditions involve excessive or deficient production of sex steroids and can alter development of primary or secondary sex characteristics in some affected infants, children, or adults. Approximately 95% of cases of CAH are due to **21-hydroxylase deficiency**.
- **Steroid 21-hydroxylase** is one of a **cytochrome P450** enzymes that is involved with the biosynthesis of the steroid hormones aldosterone and cortisol.

Congenital adrenal hyperplasia

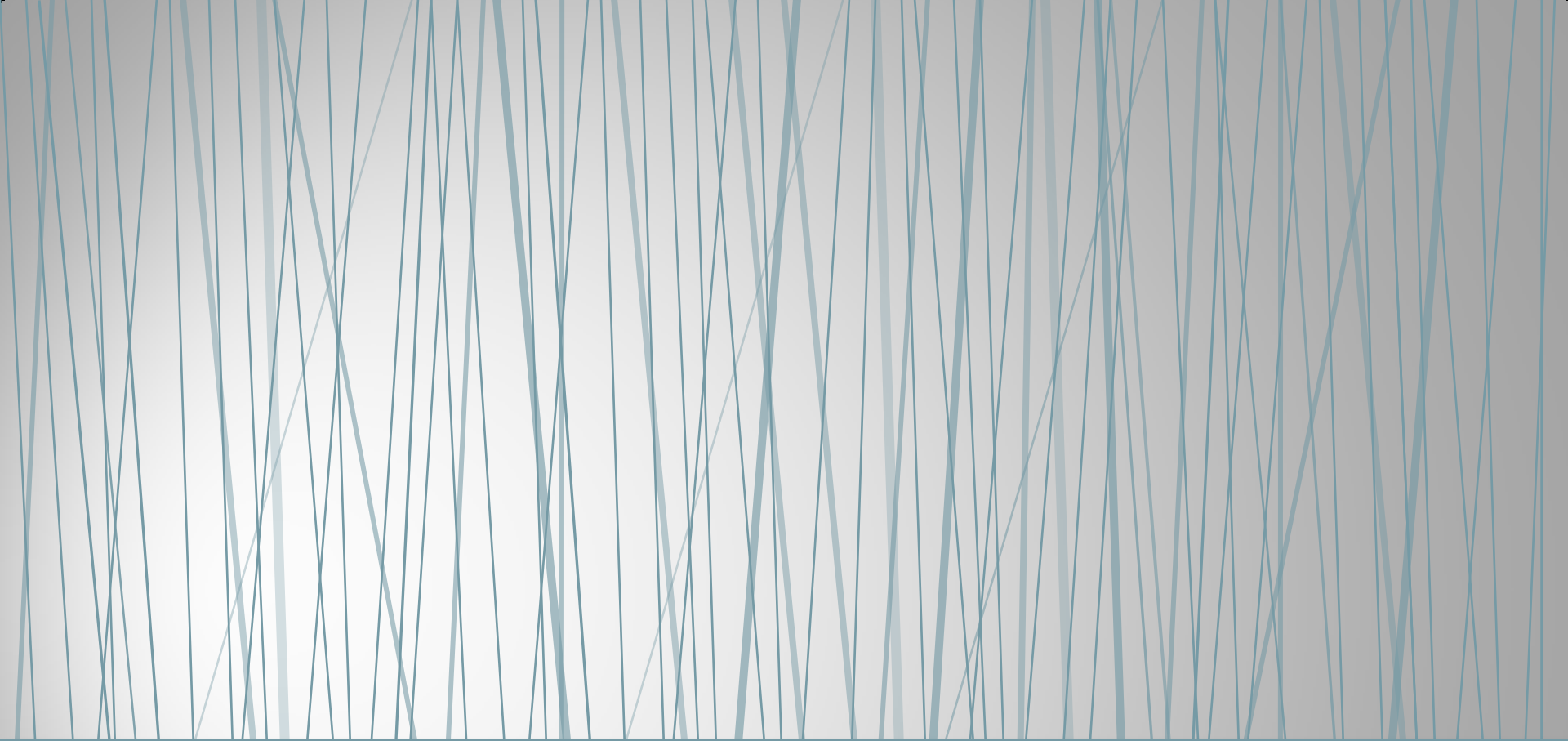
Treatment:

- Supplying enough glucocorticoid to reduce hyperplasia and overproduction of androgens or mineralocorticoids.
- Providing replacement mineralocorticoid and extra salt.
- Providing replacement testosterone or estrogen at puberty.

Diet:

- Patients with congenital adrenal hyperplasia should be on an unrestricted diet.
- Patients should have ample access to salt because salt wasting.
- Infants who have salt wasting generally benefit from supplementation with NaCl (2-4 g/d) added to their formula.
- Caloric intake may need to be monitored and restricted if excess weight gain occurs because glucocorticoids stimulate appetite.

Activity: restriction is not necessary if appropriate glucocorticoid.

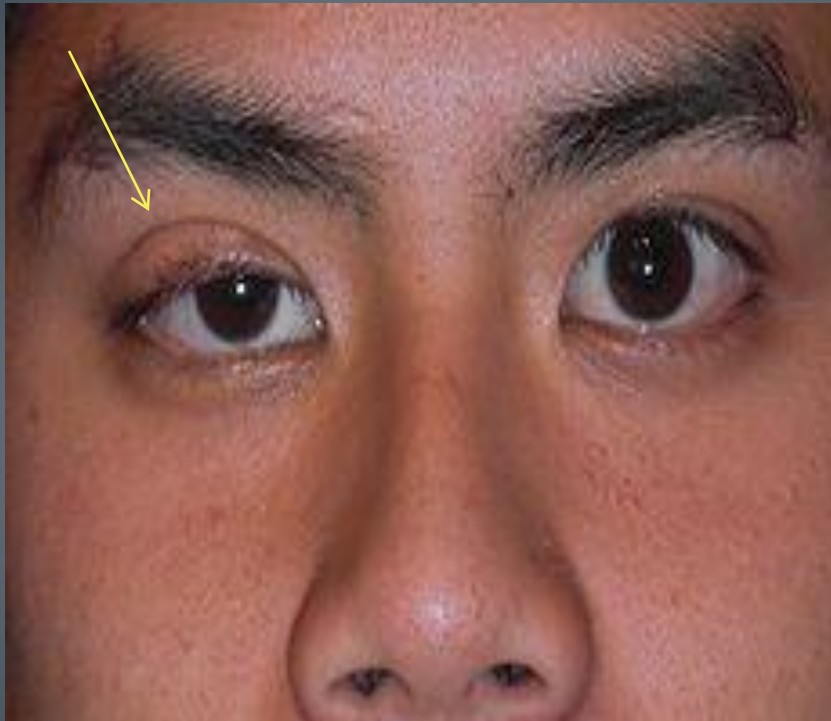
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Disorders of mitochondrial function

Kearns-Sayre Syndrome

- (KSS) is a mitochondrial myopathy with a typical onset before 20 years of age.
- KSS is a more severe syndromic variant of **chronic progressive external ophthalmoplegia** (CPEO), a syndrome that is characterized by isolated involvement of the muscles controlling eyelid movement and those controlling eye movement (extra-ocular muscles). This results in ptosis (drooping upper eyelid) and ophthalmoplegia respectively.
- KSS involves cardiac conduction abnormalities.
- Other areas of involvement can include cerebellar ataxia, deafness, diabetes mellitus, growth hormone deficiency, hypoparathyroidism, or other endocrinopathies.

Demonstrated images of KSS



Eyelid ptosis



**Chronic progressive external
ophthalmoplegia**

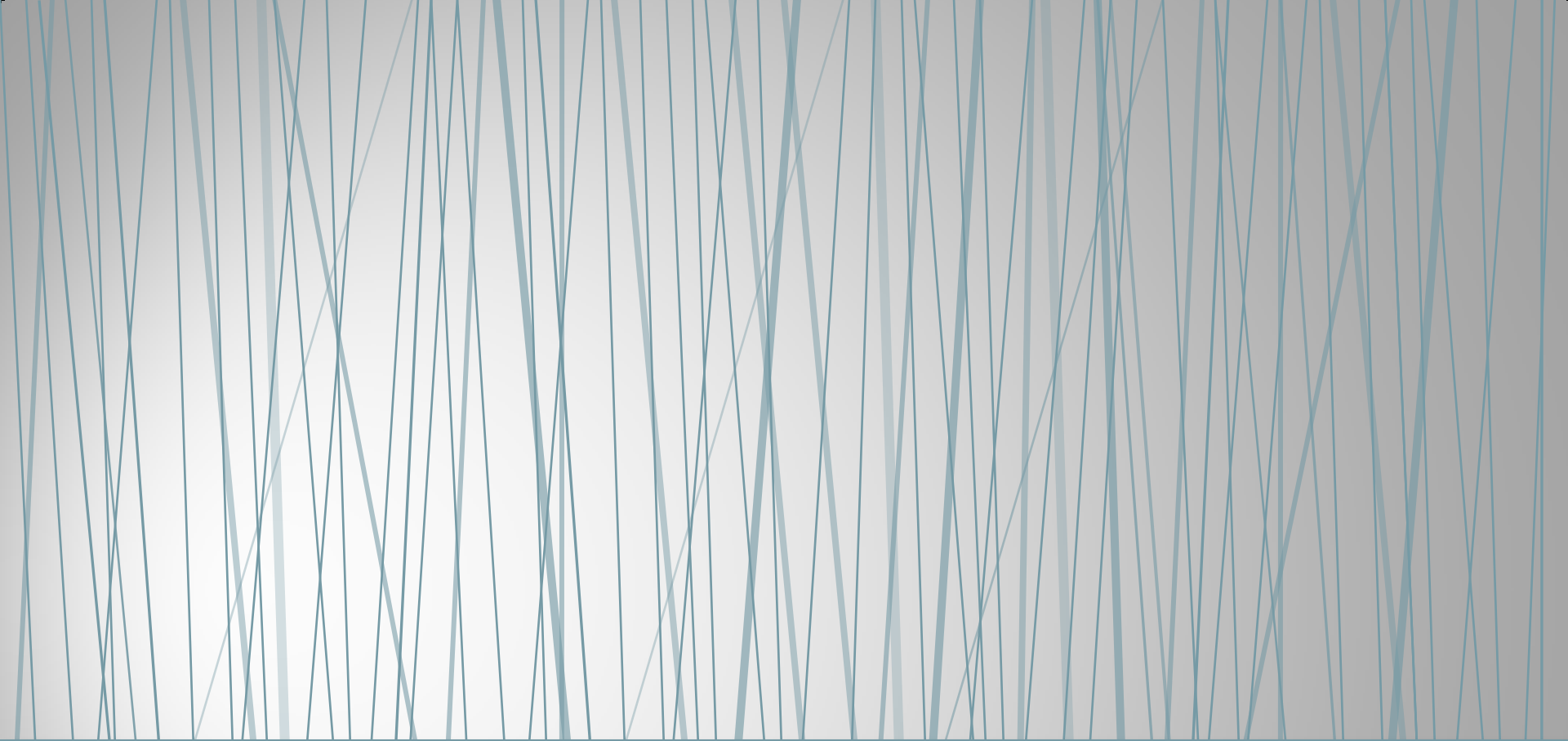
Kearns-Sayre Syndrome

- **Treatment:** Currently there is no curative treatment for KSS.
- One study described a patient with KSS who had reduced serum levels of coenzyme Q10. Administration of 60–120 mg of Coenzyme Q10 for 3 months resulted in normalization of lactate and pyruvate levels, improvement of previously diagnosed **first degree AV block**, and **improvement of ocular movements****.

Foods rich in Co Q10: beef liver, sesame & cotton seed oil, sardines, eggs, garlic and sweet potatoes.

- Screening for endocrinologic disorders should be performed, including measuring **serum glucose levels**, **thyroid function tests**, **calcium and magnesium levels**, and **serum electrolyte levels**.

** Ogasahara, S et al. (1985) "Improvement of abnormal pyruvate metabolism and cardiac conduction defect with coenzyme Q(10) in Kearns-Sayre syndrome." Neurology 35: 372-377. PubMed ID : 3974895

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Disorders of peroxisomal function

Zellweger Syndrome

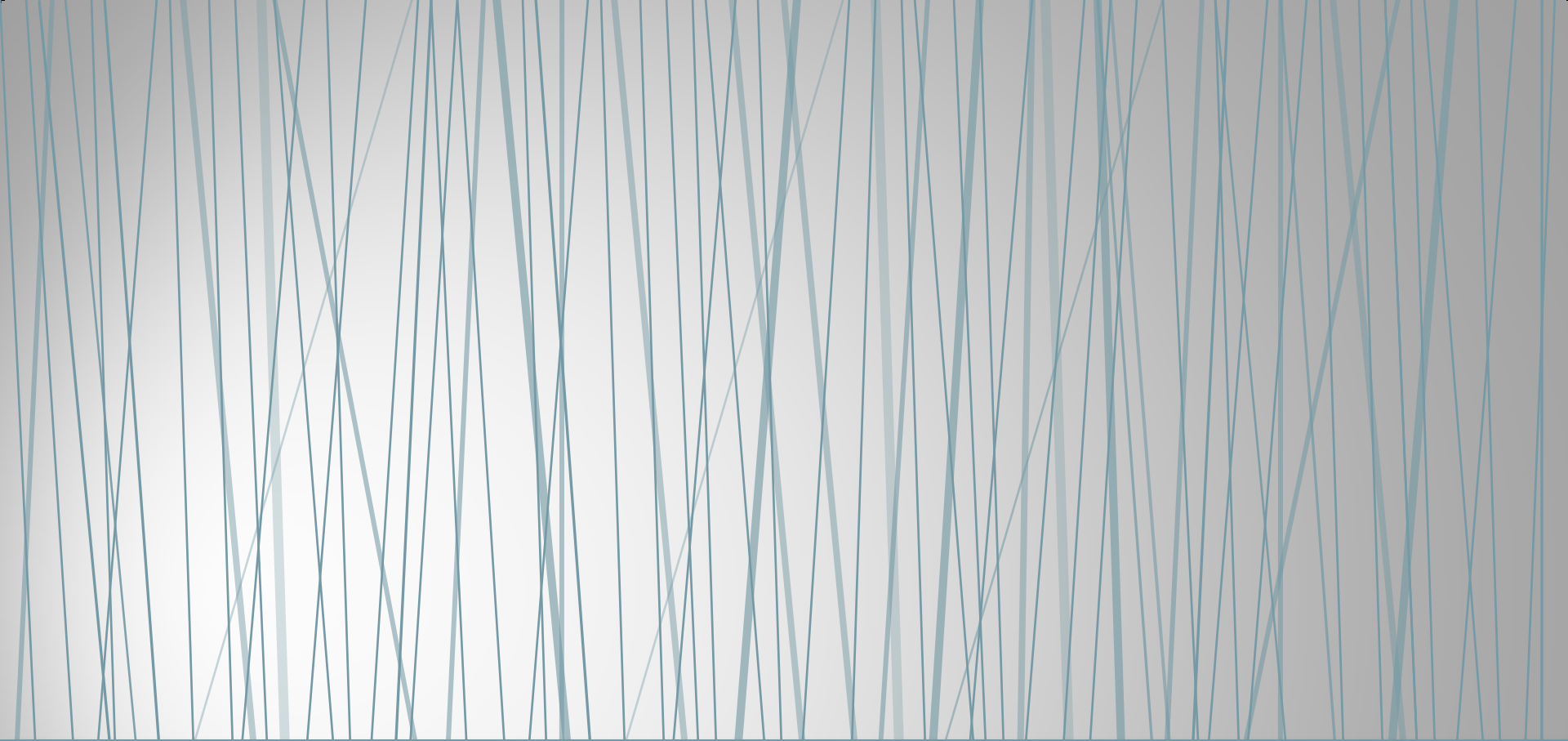
- Zellweger syndrome, also called **cerebrohepatorenal syndrome** is a rare, congenital disorder (present at birth), characterized by the reduction or absence of **peroxisomes** in the cells of the **liver, kidneys, and brain**.
- Peroxisomes contain oxidative enzymes, such as catalase, D-amino acid oxidase, and uric acid oxidase.
- It is characterized by an individual's inability to beta-oxidize **very-long chain fatty acids** in the peroxisomes of the cell.

The most features include

- 1- An enlarged **liver**, high levels of iron and copper in the blood stream, and vision disturbances.
- 2- Symptoms at birth may include a lack of muscle tone, and glaucoma.
- 3- Mental retardation, and an inability to suck and/or swallow.
- 4- Jaundice and gastrointestinal bleeding may also occur.

Zellweger Syndrome

- **Treatment:**
- Treatment of Zellweger syndrome is primarily symptomatic and supportive.
- Vitamin K may be needed to avoid abnormal bleeding.
- DHA is an essential fatty acid, which is deficient in patients with Zellweger syndrome. Improvement has been reported in some patients.
- Actually; there is no cure for Zellweger syndrome and patient will die at first year of life.

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Lysosomal storage disorders

Gaucher's disease

- **Gaucher's disease** is a genetic disease in which a fatty substance accumulates in cells and certain organs.
- It is caused by a hereditary deficiency of the enzyme **glucocerebrosidase**. The enzyme acts on a fatty substance **glucocerebroside** (also known as glucosylceramide).
- When the enzyme is defective, glucocerebroside accumulates, particularly in white blood cells (mono & lymphocyte).
- Glucocerebroside can collect in the spleen, liver, kidneys, lungs, brain and bone marrow.

Sign and Symptoms:

- Painless hepatomegaly, splenomegaly, mental retardation, and rapid and premature destruction of blood cells, leading to anemia.

Gaucher's disease

Treatment:

- The enzyme replacement therapy is essential for the treatment.
- Osteoporosis can be reduced by Vit D.
- Gaucher patients have increased caloric requirements because they have higher-than-normal metabolism.
- Despite the need for more food, patients with pronounced liver and/or spleen enlargement can frequently have a suppressed appetite. The enlarged organs leave little room in the body cavity for a full stomach, so patients often report a sensation of feeling full, even after having only a few bites of food.
- Minerals or vitamins specially B12 are recommended..

Inborn errors of metabolism

PART-II

Galactosemia

- Galactosemia is an inherited disorder characterized by an inability of the body to utilize galactose.
- Galactosemia means "galactose in the blood".
- The main source of galactose in the diet is milk products.
- The deficient enzyme that is responsible of galactosemia is called **galactose-1-phosphate uridyl transferase (GALT)**. The GALT enzyme enables the body to break down galactose into glucose for energy.
- Galactosemia is treated by removing foods that contain galactose from the diet. Untreated galactosemia will result in a harmful build-up of galactose and galactose-1-phosphate in the bloodstream and body tissues.

Galactosemia

- Infants with unrecognized galactosemia usually have problems with feeding and do not grow as they should.
- If galactosemia is not treated, infants can develop cataracts, liver disease, kidney problems, brain damage, and in some cases, can lead to death.

Diet:

- The diet should allow most protein-containing foods other than milk and milk products.
- Lactose is often used as a filler or inactive ingredient in medicines, and might not be listed on the package.

Galactosemia

Some foods contain galactose and are unacceptable:

Butter	Buttermilk and solids
Calcium caseinate	Casein
Nonfat milk	Cream
Dry milk and milk protein	beans
Hydrolyzed protein made from casein	Ice cream
Lactalbumin (milk albuminate)	Lactose
Milk and milk solids	Milk chocolate
Nonfat dry milk & solids	Cheese
Organ meats (liver, heart, etc.)	Sherbet
Sodium caseinate	Sour cream
Whey مصل اللبن and whey solids	Yogurt

- **Sherbet:** Traditional cold drink prepared of species of cherries, rose, licorice or Hibiscus with **diary** products.

Galactosemia

- **Foods with more than 10 mg Galactose/100 gram of food:**

Tomato	23
Date	11
Papaya	29
Bell Pepper	10
Watermelon	15

- **Foods with 5-10 mg Galactose/100 gram of food:**
- Apricot, Avocado, Cabbage, Cantaloupe, Cauliflower, Celery, Sweet corn, Cucumber, Eggplant, Green grapes, Grapefruit, Kale, Lettuce, Oranges, Peas, White potato, Radish Spinach, Turnip ,Apple ,Banana ,Broccoli Carrot, Kiwi, Green onion, Yellow onion, Pears, Sweet potato, Pumpkin.

Hereditary fructose intolerance

- **Is a fructose poisoning** is a hereditary condition caused by a deficiency of liver enzymes that metabolise fructose.
- Deficiency of **Fructose 1-Phosphate Aldolase** or **Aldolase B** enzyme which lead to accumulate Fructose-1-phosphate in blood.
- Aldolase-B, converts F-1-ph to Dihydroxyacetone phosphate and glyceraldehyde. (Acts in **Glycolysis** and **Gluconeogenesis**).
- Exclusive breastfeeding baby remain without symptoms.
- Symptoms include vomiting, hypoglycemia, failure to thrive, cachexia, hepatomegaly, jaundice, coagulopathy, coma, and severe metabolic acidosis (due to lactic acidosis).

Treatment: fructose and sucrose free diets.

Hereditary fructose intolerance

Avoid foods that contain:

- Fructose
- High-fructose corn syrup
- Table sugar (sucrose)
- Confectioner's sugar or powdered sugar
- Fruit and fruit juices
- Honey
- Regular sodas
- Flavored water
- Sorbitol
- Sports drinks
- Sweetened milk or sweetened milk beverages
- Molasses

Fruits And Fruit Juices:

Dates - 32 grams
Figs - 29.6 grams
Dried peaches - 13.5 grams
Dried apricots - 12.5 grams
Grapes - 8 grams
Pears - 6.2 grams
Apples - 6 grams
Apple juice - 5.6 grams
Mango - 5.5 grams
Cherries - 5.3 grams
Bananas - 4.85 grams
Kiwi Fruit - 4.3 grams
Watermelon - 3.35 grams
Orange Juice - 2.7
Strawberries - 2.4 grams
Oranges - 2.2 grams
Pineapple - 2.0 grams

Methemoglobinemia

- Is a disorder characterized by the presence of a higher level of methemoglobin in the blood.
- Methemoglobin is an oxidized form of hemoglobin, {the iron within hemoglobin is oxidized from the ferrous (Fe^{2+}) state to the ferric (Fe^{3+}) state}, that has **no affinity** for oxygen, resulting in no oxygen delivery to the tissue, so hypoxia can occur.
- **Clinically, this condition causes cyanosis.**
- The major cause of inborn is glucose-6-phosphate dehydrogenase [G6PD] deficiency and cytochrome b5 oxidase deficiency) or severe acidosis, which impairs the function of cytochrome b5 oxidase.
- This is particularly evident in young infants with diarrhea, in whom excessive stool bicarbonate loss leads to metabolic acidosis.

Methemoglobinemia

- Ascorbic acid is an antioxidant that may also be administered in patients with methemoglobin levels of more than 30%.
- Oral ascorbic acid (200-500 mg) has been found to be partially effective, some authors recommend using higher doses of up to 1000 mg/d.
- Some vegetables (beets البنجر, spinach, and carrots) are high in nitrite content, may need to be avoided in susceptible patients as well as contaminated water of nitrates and nitrites.

Food Sources of Vit C

(adapted from world healthiest food; WHFood's)

Food	Serving size	Calories	Vit-C (mg)
Bell peppers, red, raw, slices	1 cup	24.8	174.8
Broccoli, steamed	1 cup	43.7	123.4
Strawberries	1 cup	43.2	81.70
Lemon juice, fresh	¼ cup	15.3	28.06
Grapefruit	½ each	36.9	46.86
Kiwifruit	1 each	46.4	57.00
Cantaloupe	1 cup	56.0	67.52
Oranges	1 each	61.6	69.70
Tomato, ripe	1 cup	37.8	34.38
Banana	1 each	108.1	10.75
Apples	1 each	81.4	7.87
Grapes	1 cup	61.6	3.68
Avocado, slices	1 cup	235	11.53

Celiac Disease

- **Celiac Disease (CD)** is a lifelong inherited autoimmune condition affecting children and adults.
- When people with CD eat foods that contain **gluten**, it creates an immune-mediated toxic reaction that causes damage to the small intestine and does not allow food to be properly absorbed. (For children, growth failure is the biggest challenge)
- Even small amounts of **gluten** in foods can affect those with CD and cause health problems.
- Damage occur to small bowel even when no symptoms present.
- **Gluten can find in wheat, barley, and rye.**
- **Sensitivity of gluten is to Gliadin portion of protein.**

early onset osteoporosis

irritable bowel syndrome

vitamin deficiencies

weight loss

abdominal pain

constipation

gas

bloating

weakness

fatigue

diarrhea

unexplained anemia

THE FACE OF CELIAC DISEASE

Celiac disease

Gluten free diet (Accepted)	Have Gluten (Not-Accepted)
• Fresh meats, fish and poultry • Most dairy products (due to Lactase deficiency, milk may be restricted) • Fruits • Vegetables • Rice • Potatoes • Gluten-free flours (soy, corn)	• Breads (bran, germ, & semolina) • Cereals • Crackers • Pasta • Cookies • Cakes and pies • Sauces

Reading food labels is important.
Actually, **No** treatment for celiac disease.

Q: What's about Oats?

